

Mathematical preliminaries

Instructor: Roberto Colomboni

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These notes review the mathematical tools that we will use throughout the course. We begin with notation and algebra, then introduce derivatives and gradients, and finally show how they can be used to identify minima.

1 Sums and products

Suppose that $x_1, \dots, x_N$ are N real numbers. We often need to add or multiply them, for example when combining information from several observations.

Summation notation abbreviates repeated addition:

$$\sum_{i=1}^N x_i := x_1 + x_2 + \dots + x_N.$$

The index i takes each integer value from the lower limit 1 to the upper limit N , telling us which term to add. For example,

$$\sum_{i=1}^3 i^2 = 1^2 + 2^2 + 3^2 = 14.$$

Product notation does the same for multiplication:

$$\prod_{i=1}^N x_i := x_1 \cdot x_2 \cdot \dots \cdot x_N, \quad \prod_{i=1}^3 i = 1 \cdot 2 \cdot 3 = 6.$$

We will usually write the limits explicitly. When the range of the index is clear from the context, however, we may omit them. For instance, if the data are $x_1, \dots, x_N$, we may write $\sum_i x_i$ instead of $\sum_{i=1}^N x_i$, or $\prod_i x_i$ instead of $\prod_{i=1}^N x_i$.

2 Exponentials and logarithms

Throughout these notes, $\log$ is the natural logarithm and $\exp(t) = e^t$. Figure 1 shows the exponential and the natural logarithm. The marked points illustrate $e^0 = 1$ and $\log 1 = 0$.

The exponential is always positive, and

$$e^s > 0 \quad (s \in \mathbb{R}), \quad e^0 = 1, \quad \log 1 = 0. \quad (1)$$

The identity $\log 1 = 0$ in (1) follows because the logarithm and exponential undo each other:

$$e^{\log x} = x \quad (x > 0), \quad \log(e^t) = t \quad (t \in \mathbb{R}).$$

For our purposes, logarithms are useful because they turn products of positive numbers into sums:

$$\log(bc) = \log b + \log c, \quad b, c > 0. \quad (2)$$

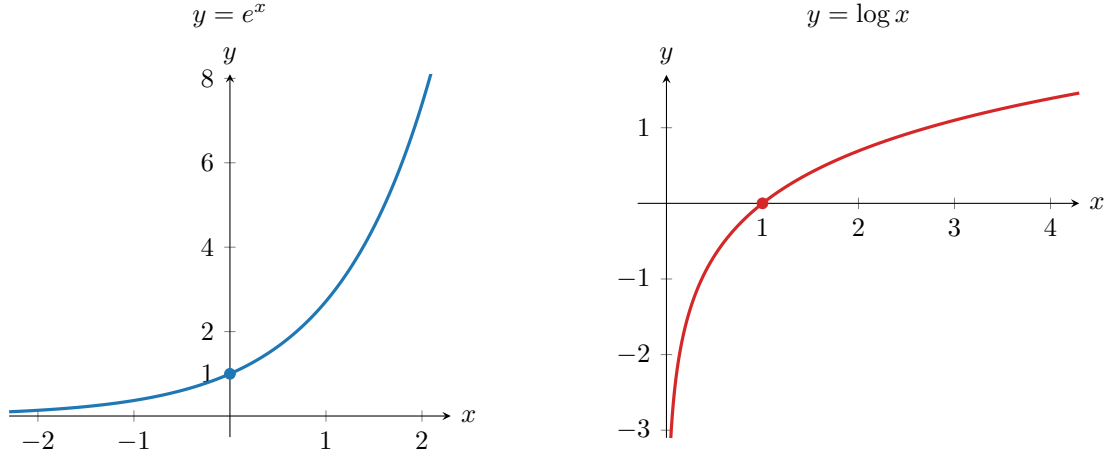


Figure 1: The exponential is defined for every real input and is strictly positive. The natural logarithm is defined for positive inputs only. Both functions are strictly increasing.

For $u, v > 0$, applying (2) to the product $(u/v)v = u$ gives $\log u = \log(u/v) + \log v$. Rearranging yields

$$\log \frac{u}{v} = \log u - \log v, \quad u, v > 0. \quad (3)$$

In particular, (3) with $u = 1$, together with (1), gives

$$\log(1/v) = -\log v, \quad v > 0.$$

For a positive integer k , applying (2) repeatedly gives

$$\log(x^k) = \log \left(\underbrace{x \cdots x}_{k \text{ factors}} \right) = \underbrace{\log x + \cdots + \log x}_{k \text{ terms}} = k \log x, \quad x > 0.$$

More generally, for positive $x_1, \dots, x_N$,

$$\log \left(\prod_{i=1}^N x_i \right) = \sum_{i=1}^N \log x_i. \quad (4)$$

This is useful for two reasons. First, sums are usually easier to differentiate than long products. Second, a product of many small positive numbers can become too small for a computer to represent as a positive number. When we need its logarithm, summing the logarithms of the factors avoids computing that tiny product first.

Order and sign. The natural logarithm is *strictly increasing*: it preserves the order of positive numbers. Equivalently,

$$u < v \iff \log u < \log v, \quad u, v > 0. \quad (5)$$

By (5), taking logarithms preserves every comparison between positive values. Maximising a positive quantity is therefore equivalent to maximising its logarithm, or to minimising its negative logarithm.

Also, comparing $p \in (0, 1)$ with 1 using (5) and (1) gives

$$\log p < \log 1 = 0, \quad \text{so} \quad -\log p > 0.$$

Thus the negative logarithm of a *probability* strictly between zero and one is positive; at probability one it is zero.

3 Vectors, matrices and index notation

To organise several quantities at once, we use vectors and matrices. Matrix multiplication provides a compact way to write sums of products.

3.1 Objects, entries and shapes

We distinguish the three types of object typographically:

Object	Notation	Example of an entry
Scalar	Ordinary type: $a \in \mathbb{R}$	a
Vector	Bold lowercase: $\mathbf{x} \in \mathbb{R}^d$	x_j
Matrix	Bold uppercase: $\mathbf{A} \in \mathbb{R}^{m \times n}$	A_{ij}

An entry of a vector or matrix is itself a scalar, so it is not bold. The space $\mathbb{R}^d$ consists of ordered lists of d real coordinates; $\mathbb{R}^{m \times n}$ consists of arrays of real numbers with m rows and n columns.

Coordinates and orientation. Writing $\mathbf{x} \in \mathbb{R}^d$ states that $\mathbf{x}$ has d real coordinates. When orientation matters, we make it explicit using $\mathbb{R}^{1 \times d}$ for rows and $\mathbb{R}^{d \times 1}$ for columns. We use *input feature vectors as rows* and *weight vectors for a single output as columns*:

$$\mathbf{x} = (x_1 \quad \cdots \quad x_d) \in \mathbb{R}^{1 \times d}, \quad \mathbf{w} = \begin{pmatrix} w_1 \\ \vdots \\ w_d \end{pmatrix} \in \mathbb{R}^{d \times 1}.$$

For a matrix, the first index always selects the row and the second selects the column:

$$\mathbf{A} = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \cdots & A_{mn} \end{pmatrix} \in \mathbb{R}^{m \times n}.$$

For example, A_{23} is the entry in the second row and third column. The row index runs from 1 to m , and the column index from 1 to n .

Addition of two matrices with the same shape and multiplication by a scalar are performed entry by entry. Specifically, the entries of $\mathbf{A} + \mathbf{B}$ and $c\mathbf{A}$ are defined by

$$(\mathbf{A} + \mathbf{B})_{ij} := A_{ij} + B_{ij}, \quad (c\mathbf{A})_{ij} := cA_{ij}.$$

Matrix multiplication, defined below, is a different operation from multiplying corresponding entries.

3.2 Transposes

Transposition exchanges rows and columns. If $\mathbf{A} \in \mathbb{R}^{m \times n}$, then the *transpose* of $\mathbf{A}$, denoted by $\mathbf{A}^\top \in \mathbb{R}^{n \times m}$, is the matrix whose entries are defined by

$$(\mathbf{A}^\top)_{ji} := A_{ij}.$$

For example,

$$\mathbf{B} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, \quad \mathbf{B}^\top = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

Applying this to the input row gives

$$\mathbf{x}^\top = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{pmatrix} \in \mathbb{R}^{d \times 1}.$$

Likewise, $\mathbf{w}^\top$ is a row when $\mathbf{w}$ is a column. The entries have not changed; their organisation has. Transposing twice restores the original object: $(\mathbf{A}^\top)^\top = \mathbf{A}$ and $(\mathbf{x}^\top)^\top = \mathbf{x}$.

3.3 Matrix–matrix multiplication

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$. Their product is the matrix $\mathbf{C} = \mathbf{AB} \in \mathbb{R}^{m \times p}$ whose entries are defined by

$$C_{ik} := \sum_{j=1}^n A_{ij} B_{jk}, \quad 1 \leq i \leq m, \quad 1 \leq k \leq p. \quad (6)$$

We first check the shapes:

$$\underbrace{\mathbf{A}}_{m \times n} \underbrace{\mathbf{B}}_{n \times p} = \underbrace{\mathbf{C}}_{m \times p}.$$

The two inner dimensions agree because each term pairs one entry of a row of $\mathbf{A}$ with one entry of a column of $\mathbf{B}$. The output retains the row index of the first matrix and the column index of the second.

There are two kinds of index in (6). The indices i, k identify the output entry and are held fixed during the sum. The index j runs through the n pairs of entries. This is why j disappears from the result while i and k remain.

3.3.1 Matrix multiplication by hand

To actually do a matrix multiplication by hand, it can be useful to use the *two-finger method*. To compute an entry C_{ik} of $\mathbf{C} = \mathbf{AB}$, move one finger left-to-right along row i of $\mathbf{A}$ and another finger top-to-bottom down column k of $\mathbf{B}$. Multiply the entries you meet in pairs and add the resulting products.

For a 2×2 product, the four entries use the following row–column pairs:

$$\begin{aligned} C_{11} : & \quad \left(\begin{array}{cc} \rightarrow & \rightarrow \end{array} \right) \left(\begin{array}{c} \downarrow \\ \downarrow \end{array} \right), & C_{12} : & \quad \left(\begin{array}{cc} \rightarrow & \rightarrow \end{array} \right) \left(\begin{array}{c} \downarrow \\ \downarrow \end{array} \right), \\ C_{21} : & \quad \left(\begin{array}{cc} & \rightarrow \end{array} \right) \left(\begin{array}{c} \downarrow \\ \downarrow \end{array} \right), & C_{22} : & \quad \left(\begin{array}{cc} & \rightarrow \end{array} \right) \left(\begin{array}{c} \downarrow \\ \downarrow \end{array} \right). \end{aligned}$$

Thus C_{11} uses the first row of $\mathbf{A}$ and the first column of $\mathbf{B}$; C_{12} uses the first row and second column; and similarly for C_{21} and C_{22} .

Worked example.

$$\begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 5 \cdot 1 + 6 \cdot 3 & 5 \cdot 2 + 6 \cdot 4 \\ 7 \cdot 1 + 8 \cdot 3 & 7 \cdot 2 + 8 \cdot 4 \end{pmatrix} = \begin{pmatrix} 23 & 34 \\ 31 & 46 \end{pmatrix}.$$

For instance, the top-right entry is $5 \cdot 2 + 6 \cdot 4 = 34$. It uses the top row of the first matrix and the right column of the second.

Multiplication is associative (when the shapes permit it): $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$. It is not generally commutative: $\mathbf{AB}$ and $\mathbf{BA}$ can differ, or only one may be defined. Exercise 9 reverses the two matrices in the example above.

A useful identity is

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top.$$

Notice that the order of the two factors is reversed.

3.4 Matrix–vector and vector–matrix multiplication

The same definition covers products involving a vector. If $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{c} \in \mathbb{R}^{n \times 1}$, then

$$\underbrace{\mathbf{v}}_{m \times 1} = \underbrace{\mathbf{A}}_{m \times n} \underbrace{\mathbf{c}}_{n \times 1}, \quad v_i = \sum_{j=1}^n A_{ij} c_j. \quad (7)$$

The result is a column vector: each of its coordinates is the sum of one row of $\mathbf{A}$ weighted by $\mathbf{c}$.

For a row vector $\mathbf{r} \in \mathbb{R}^{1 \times m}$, multiplication on the other side gives

$$\underbrace{\mathbf{s}}_{1 \times n} = \underbrace{\mathbf{r}}_{1 \times m} \underbrace{\mathbf{A}}_{m \times n}, \quad s_j = \sum_{i=1}^m r_i A_{ij}. \quad (8)$$

The result is a row vector. Both products are instances of the matrix multiplication rule above.

3.5 Inner products and outer products

The *inner product* of two lists of d real coordinates is the sum of the coordinatewise products. With an input row $\mathbf{x}$ and a weight column $\mathbf{w}$, it is written

$$\mathbf{x}\mathbf{w} = \sum_{j=1}^d x_j w_j. \quad (9)$$

The shapes are $(1 \times d)(d \times 1)$, giving a 1×1 matrix, which we identify with its scalar entry.

For two rows $\mathbf{x}, \mathbf{r}$, the same coordinate formula is written $\mathbf{x}\mathbf{r}^\top$; for two columns $\mathbf{u}, \mathbf{v}$, it is $\mathbf{u}^\top \mathbf{v}$. In particular,

$$\mathbf{x}\mathbf{x}^\top = \sum_{j=1}^d x_j^2, \quad \mathbf{w}^\top \mathbf{w} = \sum_{j=1}^d w_j^2.$$

The square root of the sum of squared coordinates is called the *Euclidean norm*, written $\|\mathbf{x}\|_2$ or $\|\mathbf{w}\|_2$.

The *outer product* is defined differently. For columns $\mathbf{u} \in \mathbb{R}^{m \times 1}$ and $\mathbf{v} \in \mathbb{R}^{p \times 1}$,

$$\underbrace{\mathbf{M}}_{m \times p} = \underbrace{\mathbf{u}}_{m \times 1} \underbrace{\mathbf{v}^\top}_{1 \times p}, \quad M_{ik} = u_i v_k. \quad (10)$$

The inner dimension is 1, so each output entry is a single product.

In particular, the outer products of a vector with itself are

$$\mathbf{w}\mathbf{w}^\top \in \mathbb{R}^{d \times d}, \quad \mathbf{x}^\top \mathbf{x} \in \mathbb{R}^{d \times d}.$$

Unlike the inner products above, these give matrices rather than scalars.

3.6 The identity matrix

The identity matrix $\mathbf{I}_n \in \mathbb{R}^{n \times n}$ has ones on its diagonal and zeros elsewhere. For example,

$$\mathbf{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

It leaves a compatible vector or matrix unchanged. For a column $\mathbf{w} \in \mathbb{R}^{n \times 1}$, a row $\mathbf{x} \in \mathbb{R}^{1 \times n}$ and a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$:

$$\mathbf{I}_n \mathbf{w} = \mathbf{w}, \quad \mathbf{x} \mathbf{I}_n = \mathbf{x}, \quad \mathbf{I}_m \mathbf{A} = \mathbf{A}, \quad \mathbf{A} \mathbf{I}_n = \mathbf{A} \quad (\mathbf{A} \in \mathbb{R}^{m \times n}).$$

The size of the identity is determined by the side on which it is multiplied. For a rectangular matrix, the identity matrix on the left need not have the same size as the identity matrix on the right.

3.7 Matrix inverses and linear systems

A square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is *invertible* if there exists a matrix $\mathbf{B} \in \mathbb{R}^{n \times n}$ such that

$$\mathbf{A}\mathbf{B} = \mathbf{B}\mathbf{A} = \mathbf{I}_n.$$

It can be shown that, if such a matrix $\mathbf{B}$ exists, then it is unique. We call it the *inverse* of $\mathbf{A}$ and denote it by $\mathbf{A}^{-1}$. Thus,

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n.$$

The inverse undoes multiplication by $\mathbf{A}$. A square matrix that is not invertible is called *singular*.

For a 2×2 matrix,

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad \text{if } ad - bc \neq 0. \quad (11)$$

The number $ad - bc$ is the determinant of this 2×2 matrix. In this case, being nonzero is exactly the condition for invertibility.

To verify the formula, multiply:

$$\mathbf{A}^{-1}\mathbf{A} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{ad - bc} \begin{pmatrix} ad - bc & db - bd \\ -ca + ac & ad - bc \end{pmatrix} = \mathbf{I}_2.$$

Multiplication in the other order also gives $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_2$.

Solving a linear system. For columns $\mathbf{u}, \mathbf{c} \in \mathbb{R}^{n \times 1}$, if $\mathbf{A}$ is invertible and $\mathbf{A}\mathbf{u} = \mathbf{c}$, then multiplying both sides on the left by $\mathbf{A}^{-1}$ gives

$$\mathbf{u} = \mathbf{I}_n \mathbf{u} = (\mathbf{A}^{-1} \mathbf{A}) \mathbf{u} = \mathbf{A}^{-1} (\mathbf{A} \mathbf{u}) = \mathbf{A}^{-1} \mathbf{c}.$$

Exercises 10 and 11 ask you to carry this out numerically.

4 The Kronecker delta

The Kronecker delta is the number

$$\delta_{ij} := \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \quad (12)$$

These are precisely the entries of the identity matrix: $(\mathbf{I}_d)_{ij} = \delta_{ij}$. We will use this notation to select coordinates and, later, when differentiating expressions involving vectors and sums.

Indicators and counts. An indicator records whether a condition A is true:

$$\mathbf{1}\{A\} := \begin{cases} 1, & \text{if } A \text{ is true,} \\ 0, & \text{if } A \text{ is false.} \end{cases} \quad (13)$$

Comparing (13) with (12), we see that $\delta_{ij} = \mathbf{1}\{i = j\}$.

A sum of indicators counts how many of their conditions hold. For instance, if y_i is the observed class label and $\hat{y}_i$ is a predicted label for observation i , then

$$\text{number of mistakes} = \sum_{i=1}^N \mathbf{1}\{\hat{y}_i \neq y_i\}. \quad (14)$$

Each incorrect prediction contributes one to (14); each correct prediction contributes zero. For example, with

$$(y_1, y_2, y_3, y_4) = (0, 1, 1, 0), \quad (\hat{y}_1, \hat{y}_2, \hat{y}_3, \hat{y}_4) = (0, 0, 1, 1),$$

the count in (14) is $0 + 1 + 0 + 1 = 2$.

4.1 Selecting one term from a sum

A Kronecker delta often simplifies a sum by selecting a single term. For example,

$$\sum_{j=1}^3 \delta_{2j} a_j = \delta_{21} a_1 + \delta_{22} a_2 + \delta_{23} a_3 = 0 \cdot a_1 + 1 \cdot a_2 + 0 \cdot a_3 = a_2.$$

For any real numbers $a_1, \dots, a_n$ and a fixed $i \in \{1, \dots, n\}$,

$$\sum_{j=1}^n \delta_{ij} a_j = a_i. \quad (15)$$

All terms except $j = i$ are multiplied by zero, and the remaining term is multiplied by one.

The identity (15) will be useful when derivatives introduce Kronecker deltas in Section 5.

5 Derivatives and gradients

5.1 Functions, values and derivatives

A real-valued function assigns a real number to each input in its domain. For example,

$$f : \mathbb{R} \longrightarrow \mathbb{R}, \quad x \longmapsto x^2$$

means that f takes a real number as input and returns its square. Here $\mathbb{R}$ is the set of real numbers. The symbol f denotes the function, while $f(x)$ denotes its value at the input x . In particular, $f(3) = 9$. We will often specify a function by a formula such as $f(x) = x^2$.

A derivative describes how that value changes when the input changes. Let f be defined on an open interval containing x_0 . Its derivative at x_0 is

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}, \quad (16)$$

provided that this limit exists as a finite real number. The quotient is the change in the output divided by the change in the input. Taking the limit gives the slope at the point.

An alternative notation for the derivative is

$$f'(x_0) = \left. \frac{d}{dx} f(x) \right|_{x=x_0}.$$

The vertical bar means that we first differentiate the function $x \mapsto f(x)$ and then evaluate the result at x_0 . Without an evaluation point, f' denotes the derivative function. In a less cumbersome way—and when it does not cause ambiguity or confusion—we use the notation

$$\frac{d}{dx} f(x)$$

to denote the function f' evaluated at x . A positive derivative means that a *sufficiently small* positive change in the input increases the output; a negative derivative means that it decreases it. A zero derivative requires more care, as we will see in Section 6.

Figure 2 gives a geometric interpretation of the difference quotient in (16). For a fixed nonzero increment h , the quotient

$$\frac{f(x_0 + h) - f(x_0)}{h}$$

is the slope of the secant line through the two points of the graph with horizontal coordinates x_0 and $x_0 + h$. As h becomes smaller, the second point moves towards the first one, and the secant line approaches the tangent line at x_0 . The derivative is the limiting slope.

A derivative gives a local linear approximation. The definition (16) also tells us how to approximate a nearby function value. If f is differentiable at x , then, for a small change h ,

$$f(x + h) \approx f(x) + f'(x)h. \quad (17)$$

The term $f'(x)h$ in (17) is the linear prediction of the change in the output. The approximation uses the tangent line at x and hence, in general, it is not an exact equality. Differentiability means that the absolute error in this approximation, divided by $|h|$, tends to zero as $h \rightarrow 0$.

For example, take $f(x) = x^2$ and expand around $x = 2$:

$$f(2 + h) = (2 + h)^2 = 4 + 4h + h^2. \quad (18)$$

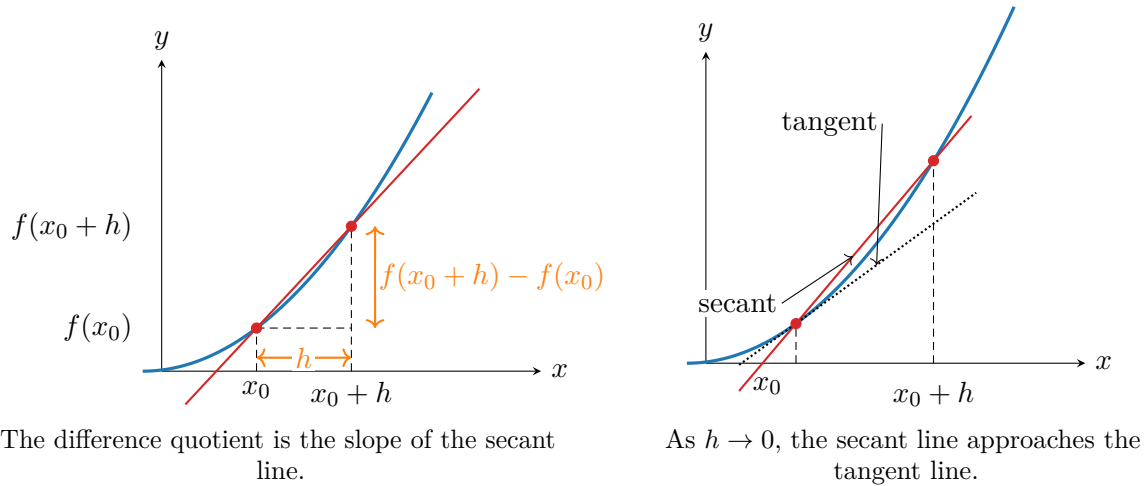


Figure 2: Geometric meaning of the derivative in (16).

Combining (18) with the derivative definition (16) gives

$$f'(2) = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0} \frac{4 + 4h + h^2 - 4}{h} = \lim_{h \rightarrow 0} (4 + h) = 4.$$

Thus (17), with $h = 0.01$, gives

$$4.0401 = f(2.01) \approx f(2) + f'(2) \cdot 0.01 = 4 + 4 \cdot 0.01 = 4.04.$$

The exact value from (18) is 4.0401, and hence the omitted term is $h^2 = 0.0001$, which becomes small relative to $|h|$ as the input change becomes small.

5.2 Standard derivatives and linear combinations

For every fixed real exponent $p \neq 0$, the power rule on $x > 0$ is

$$\frac{d}{dx} x^p = p x^{p-1}. \quad (19)$$

For positive integer powers the rule also holds for negative inputs and at zero. For negative integer powers it holds for every $x \neq 0$. Fractional powers need their domains checked: for example, $\sqrt{x}$ is real-valued for $x \geq 0$, but the derivative below is well-defined only for $x > 0$.

The examples we will use include

$$\begin{aligned} \frac{d}{dx} x^2 &= 2x, & \frac{d}{dx} x^5 &= 5x^4, \\ \frac{d}{dx} x^{-3} &= -3x^{-4} \quad (x \neq 0), & \frac{d}{dx} \sqrt{x} &= \frac{d}{dx} x^{1/2} = \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}} \quad (x > 0). \end{aligned}$$

A constant function has derivative zero, and the identity function has derivative one:

$$\frac{d}{dx} c = 0, \quad \frac{d}{dx} x = 1.$$

Two other derivative rules that we will use repeatedly are

$$\frac{d}{dx} e^x = e^x \quad (x \in \mathbb{R}), \quad \frac{d}{dx} \log x = \frac{1}{x} \quad (x > 0). \quad (20)$$

These are the derivatives of the exponential and natural logarithm introduced in Section 2.

Differentiation is linear. If u and v are differentiable and a, b are constants with respect to x , then

$$\frac{d}{dx}(a u(x) + b v(x)) = a \frac{d}{dx}u(x) + b \frac{d}{dx}v(x). \quad (21)$$

The same rule applies to any finite number of terms.

Worked example. For $x > 0$, let

$$f(x) = 3x^4 + 2x^{-1/2} + x^{-2}.$$

We differentiate each term and retain its coefficient:

$$f'(x) = 3(4x^3) + 2\left(-\frac{1}{2}x^{-3/2}\right) - 2x^{-3} = 12x^3 - x^{-3/2} - 2x^{-3}.$$

5.3 The chain rule

Many functions are evaluated in stages. We might first add one to an input and then cube the result. Define

$$u(x) := x + 1, \quad v(s) := s^3, \quad F(x) := v(u(x)).$$

The function u acts on x , the function v acts on the intermediate value $s = u(x)$, and F is the resulting *composition*. The function F , which is the composition of v with u , is also written as $F = v \circ u$.

Suppose that u is differentiable at x and v is differentiable at $u(x)$. The *chain rule* states that $F = v \circ u$ is differentiable at x and

$$F'(x) = v'(u(x)) u'(x), \quad (22)$$

or, more formally,

$$\frac{d}{dx}F(x) = \frac{d}{ds}v(s) \Big|_{s=u(x)} \frac{d}{dx}u(x). \quad (23)$$

The outer derivative is evaluated at $u(x)$, the value received by the outer function.

Worked example. For the functions above,

$$u'(x) = 1, \quad v'(s) = 3s^2.$$

Evaluating the derivative of the outer function at $s = u(x) = x + 1$ gives

$$F'(x) = 3(u(x))^2 \cdot 1 = 3(x + 1)^2.$$

Using the chain rule we did not need to expand the cube: exercises 2 and 3 ask you to compare the two methods on a different numerical example.

Writing $s = u(x)$ and $y = v(s)$, the same chain rule is commonly abbreviated as

$$\frac{dy}{dx} = \frac{dy}{ds} \frac{ds}{dx}.$$

Here the three derivatives mean $F'(x)$, $v'(u(x))$ and $u'(x)$, respectively. The notation makes it look as though the ds terms cancel, as they would in a product of fractions. However, dy/ds

must be read as a whole: it is a single symbol denoting a derivative, not the ratio of two numbers dy and dx that can be manipulated separately. The apparent cancellation is therefore a useful way to remember the chain rule, not a proof.

Later, we will see how *backpropagation* uses repeated applications of the chain rule to compute derivatives efficiently. Exercise 15 combines the derivative rules (20) with the chain rule (22) to differentiate a function that will reappear in classification.

5.4 The product rule

If $F(x) := u(x)v(x)$, both factors change when x changes. The *product rule* accounts for both contributions. Specifically, it states that if u and v are differentiable at x , then F is also differentiable at x and

$$F'(x) = u'(x)v(x) + u(x)v'(x), \quad (24)$$

or, equivalently,

$$F'(x) = \frac{du}{dx}(x) \cdot v(x) + u(x) \cdot \frac{dv}{dx}(x).$$

Worked example. Consider

$$F(x) = (x + x^2 + x^3)(2x + 3x^2 + x^3).$$

We could expand the product first, but it is shorter to define

$$u(x) = x + x^2 + x^3, \quad v(x) = 2x + 3x^2 + x^3.$$

Their derivatives are

$$u'(x) = 1 + 2x + 3x^2, \quad v'(x) = 2 + 6x + 3x^2.$$

Substitution in (24) gives

$$F'(x) = (1 + 2x + 3x^2)(2x + 3x^2 + x^3) + (x + x^2 + x^3)(2 + 6x + 3x^2).$$

A function may depend on several arguments. To differentiate with respect to one of them, we hold the others fixed.

5.5 Partial derivatives: varying one argument

Consider

$$F : \mathbb{R}^2 \longrightarrow \mathbb{R}, \quad F(a, b) = a^2 + 3ab + b^2.$$

To study how the first argument affects the output at $(1, 2)$, fix $b = 2$. This gives a function of one variable:

$$g(a) = F(a, 2) = a^2 + 6a + 4, \quad g'(1) = 8.$$

We call this the *partial derivative of F with respect to its first argument at $(1, 2)$* , and write $\partial_1 F(1, 2) = 8$. The subscript 1 specifies which argument we vary.

Similarly, fixing $a = 1$ gives

$$h(b) = F(1, b) = 1 + 3b + b^2, \quad h'(2) = 7,$$

so $\partial_2 F(1, 2) = 7$. A partial derivative is an ordinary derivative after the other arguments have been fixed.

At a general point, treat b as constant when differentiating with respect to a , and a as constant when differentiating with respect to b :

$$\partial_1 F(a, b) = 2a + 3b, \quad \partial_2 F(a, b) = 3a + 2b. \quad (25)$$

The answers can depend on both arguments: for instance, the slope of $a \mapsto 3ab$ is $3b$, which depends on the fixed value chosen for b .

Because we have named the arguments a, b , we can also write

$$\partial_1 F(a, b) = \frac{\partial}{\partial a} F(a, b), \quad \partial_2 F(a, b) = \frac{\partial}{\partial b} F(a, b).$$

The notation $(\partial F / \partial a)(1, 2)$ means the same as $\partial_1 F(1, 2)$: first differentiate with respect to the argument named a , then evaluate at $(1, 2)$.

In terms of the limit definition, the first partial derivative is

$$\partial_1 F(a_0, b_0) = \lim_{\varepsilon \rightarrow 0} \frac{F(a_0 + \varepsilon, b_0) - F(a_0, b_0)}{\varepsilon}, \quad (26)$$

when this limit exists. Only the first argument changes; the definition for the second argument works in the same way.

5.6 Gradients: collecting the partial derivatives

Let Q be a scalar-valued function of a column $\mathbf{w} = (w_1, \dots, w_d)^\top$. When all its partial derivatives exist at $\mathbf{w}$, the *gradient* collects them into a vector:

$$\nabla Q(\mathbf{w}) = \begin{pmatrix} \partial_1 Q(\mathbf{w}) \\ \vdots \\ \partial_d Q(\mathbf{w}) \end{pmatrix}, \quad \partial_j Q(\mathbf{w}) = \left(\frac{\partial Q}{\partial w_j} \right) (\mathbf{w}). \quad (27)$$

For example, take $d = 2$ and define $Q(\mathbf{w}) = F(w_1, w_2)$ using the function above. Then

$$\nabla Q(\mathbf{w}) = \begin{pmatrix} 2w_1 + 3w_2 \\ 3w_1 + 2w_2 \end{pmatrix}, \quad \nabla Q\left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}\right) = \begin{pmatrix} 8 \\ 7 \end{pmatrix}. \quad (28)$$

All entries are evaluated at the same parameter vector.

Our convention is that **a gradient has the same shape as the argument being differentiated**: a column for a column argument, a row for a row argument, and a matrix for a matrix argument. When a function has several argument blocks, a subscript identifies the block. For example, $\nabla_{\mathbf{w}} L(\mathbf{w}, b)$ collects the weight derivatives while holding b fixed.

Differentiability: a local linear approximation. The scalar approximation (17) has an analogue for a function of several parameters. Let $Q : \mathbb{R}^d \rightarrow \mathbb{R}$, let $\mathbf{w} \in \mathbb{R}^{d \times 1}$, and let $\Delta \in \mathbb{R}^{d \times 1}$ be a small change to $\mathbf{w}$.

We say that Q is *differentiable* at $\mathbf{w}$ if there exists a vector $\mathbf{g} \in \mathbb{R}^{d \times 1}$ such that

$$Q(\mathbf{w} + \Delta) = Q(\mathbf{w}) + \mathbf{g}^\top \Delta + \text{error}(\Delta), \quad \lim_{\substack{\Delta \rightarrow \mathbf{0} \\ \Delta \neq \mathbf{0}}} \frac{|\text{error}(\Delta)|}{\|\Delta\|_2} = 0.$$

The vector $\mathbf{g}$ is fixed as Δ varies. The limit requires the error to become negligible compared with the size of the increment, regardless of the direction from which Δ approaches $\mathbf{0}$.

One can show that, if such a vector exists, then all partial derivatives of Q at $\mathbf{w}$ exist and

$$g_j = \partial_j Q(\mathbf{w}) \quad \text{for every } j = 1, \dots, d, \quad \text{so} \quad \mathbf{g} = \nabla Q(\mathbf{w}).$$

In particular, this vector is unique. This follows by considering increments that vary only one coordinate at a time.

The converse is false: the existence of all partial derivatives at $\mathbf{w}$, and hence of the gradient $\nabla Q(\mathbf{w})$, does not by itself guarantee differentiability there. A sufficient condition is that all partial derivatives **exist and are continuous in a neighbourhood of $\mathbf{w}$** . For example, every polynomial in finitely many variables is differentiable everywhere: its partial derivatives are themselves polynomials and hence continuous.

For a differentiable function Q , we therefore obtain the local linear approximation

$$Q(\mathbf{w} + \Delta) \approx Q(\mathbf{w}) + (\nabla Q(\mathbf{w}))^\top \Delta = Q(\mathbf{w}) + \sum_{j=1}^d \frac{\partial Q}{\partial w_j}(\mathbf{w}) \Delta_j. \quad (29)$$

The inner product in (29) adds one contribution per coordinate: its partial derivative times its change. It is a scalar, since its shape is $(1 \times d)(d \times 1)$.

For the function $Q(\mathbf{w}) = w_1^2 + 3w_1w_2 + w_2^2$, use the gradient (28) at $\mathbf{w} = (1, 2)^\top$, where $Q(\mathbf{w}) = 11$. Taking $\Delta = (0.01, -0.02)^\top$ in (29) gives

$$Q\left(\begin{pmatrix} 1.01 \\ 1.98 \end{pmatrix}\right) \approx 11 + 8 \cdot 0.01 + 7 \cdot (-0.02) = 10.94.$$

Direct substitution gives the exact value 10.9399.

5.7 Using the chain rule for one partial derivative

For a multivariate function, the scalar chain rule still applies when we vary one variable at a time and hold the others fixed. For example, define

$$u(a, b) := a + 2b - 3, \quad q(s) := s^2, \quad R(a, b) := q(u(a, b)) = (a + 2b - 3)^2.$$

Holding b fixed leaves a function of a alone. The inner derivative is 1 and the outer derivative is $q'(s) = 2s$, evaluated at $s = u(a, b)$. Thus

$$\begin{aligned} \partial_1 R(a, b) &= q'(u(a, b)) \partial_1 u(a, b) = 2(a + 2b - 3) \cdot 1, \\ \partial_2 R(a, b) &= q'(u(a, b)) \partial_2 u(a, b) = 2(a + 2b - 3) \cdot 2. \end{aligned}$$

In the second line, we instead hold a fixed and differentiate with respect to b . Both calculations use the ordinary chain rule from Section 5.3.

When differentiating with respect to one argument, the other arguments are fixed, but an intermediate quantity such as $u(a, b)$ still changes if it depends on the argument being varied.

5.8 Differentiating vector components

Consider the identity function $\iota(\mathbf{r}) = \mathbf{r}$, where $\mathbf{r} = (r_1, \dots, r_d)$ is an input row. Its i th scalar output function is $\iota_i(\mathbf{r}) = r_i$. Varying coordinate r_j gives

$$\partial_j \iota_i(\mathbf{r}) = \delta_{ij}. \quad (30)$$

Only output j changes: its derivative is one, and the other derivatives are zero.

The same coordinate calculation is commonly abbreviated as

$$\frac{\partial r_i}{\partial r_j} = \delta_{ij}.$$

Here r_i stands for the coordinate function $\mathbf{r} \mapsto r_i$, and the denominator selects the independently variable coordinate r_j . For a row with three coordinates, differentiating the identity function one output at a time gives

$$\frac{\partial \mathbf{r}}{\partial r_2} = \begin{pmatrix} \frac{\partial r_1}{\partial r_2} & \frac{\partial r_2}{\partial r_2} & \frac{\partial r_3}{\partial r_2} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}.$$

This varies one input coordinate and lists the derivatives of all output components. The same component rule applies to column vectors such as $\mathbf{w}$.

For example, with fixed coefficients $x_1, \dots, x_d$, define $s(\mathbf{w}) = \sum_{j=1}^d x_j w_j$. Differentiating each term of this finite sum and keeping the coefficients fixed gives

$$\frac{\partial s}{\partial w_k}(\mathbf{w}) = \sum_{j=1}^d x_j \frac{\partial w_j}{\partial w_k} = \sum_{j=1}^d x_j \delta_{jk} = x_k.$$

Exercise 12 combines these ideas: write a matrix–vector product as a sum, differentiate with respect to one weight, and simplify using the Kronecker delta.

6 Using derivatives and gradients to find minima

6.1 One parameter

Now that we know how to compute derivatives, we turn to one of their main applications: finding where a function attains its minimum.

Let

$$\mathcal{L} : \mathbb{R} \longrightarrow \mathbb{R}$$

be a function of one real number a . We want to find values of a for which $\mathcal{L}(a)$ is as small as possible. We call $\mathcal{L}$ the *objective function*.

A *global minimiser* a^* satisfies

$$\mathcal{L}(a^*) \leq \mathcal{L}(a) \quad \text{for every } a \in \mathbb{R}.$$

A *local minimiser* satisfies the same inequality for all a sufficiently close to a^* .

Derivatives help us identify possible minimisers. If $\mathcal{L}$ is differentiable at a local minimiser a^* , *Fermat's theorem* states that

$$\mathcal{L}'(a^*) = 0. \quad (31)$$

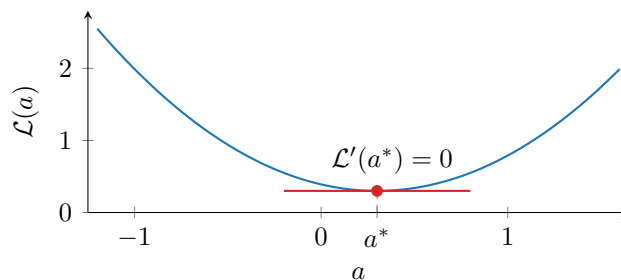


Figure 3: A schematic example: $\mathcal{L}(a) = (a - 0.3)^2 + 0.3$ has its minimum at $a^* = 0.3$, where the tangent is horizontal.

Geometrically, the tangent to the graph is horizontal there.

A point where the derivative is zero is called a *stationary point*. A stationary point need not be a minimiser: at zero, the functions a^2 , $-a^2$ and a^3 all have derivative zero, but they have a minimum, a maximum, and neither, respectively.

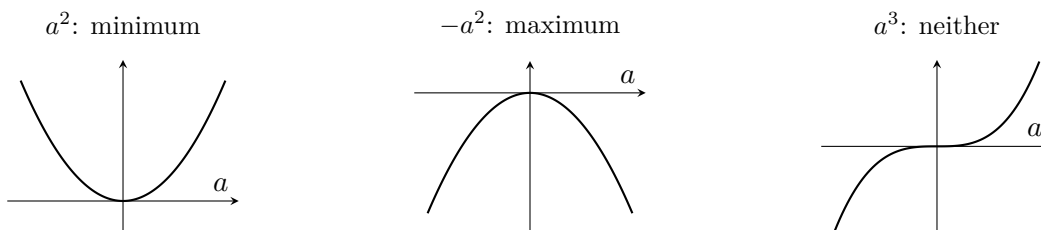


Figure 4: The derivative at zero is zero in all three examples. That fact alone does not classify the stationary point.

Worked example. Consider the quadratic objective

$$F(a) = a^2 + 5a - 3, \quad a \in \mathbb{R}.$$

The coefficient of a^2 is 1, which is positive, so the graph of F is an upward-opening parabola. Therefore, F has a global minimum. We now use its derivative to find where this minimum is attained.

Since F is differentiable everywhere on $\mathbb{R}$, Fermat's theorem implies that any point at which it attains its minimum must be a stationary point. We therefore look for the solutions of $F'(a) = 0$.

Differentiating gives

$$F'(a) = 2a + 5.$$

Hence

$$F'(a) = 0 \iff 2a + 5 = 0 \iff a = -\frac{5}{2}.$$

We see that there is only one stationary point. Since a global minimum exists and must occur at a stationary point, we conclude that

$$a^* = -\frac{5}{2}$$

is the unique global minimiser.

To find the minimum value, we substitute a^* into F :

$$F(a^*) = \left(-\frac{5}{2}\right)^2 + 5\left(-\frac{5}{2}\right) - 3 = \frac{25}{4} - \frac{25}{2} - 3 = -\frac{37}{4}.$$

The minimum value and where it is attained. In the previous example, we found two different numbers:

$$a^* = -\frac{5}{2}, \quad F(a^*) = -\frac{37}{4}.$$

The first is the parameter value that we should choose. The second is the value of the objective at that choice.

The notation $\min$ answers the question: *what is the smallest value of the objective?* In our example,

$$\min_{a \in \mathbb{R}} F(a) = -\frac{37}{4}.$$

The condition $a \in \mathbb{R}$ underneath $\min$ tells us that we are allowed to choose any real value of a .

The notation argmin answers a different question: *which values of the parameter attain that minimum?* We collect all such values in a set. In our example,

$$\operatorname{argmin}_{a \in \mathbb{R}} F(a) = \left\{-\frac{5}{2}\right\}.$$

There is only one minimiser, so this set contains only one element, i.e., it is a singleton.

More generally, suppose that the parameter is required to belong to a set Θ of allowed values. We write

$$a^* \in \operatorname{argmin}_{a \in \Theta} F(a)$$

to say that a^* is one of the allowed parameter values at which F attains its minimum over Θ .

Maximum values, maximising indices and ties. The corresponding notation for maximisation is $\max$ and argmax . The first gives the largest value, when it exists; the second gives the set of arguments at which that value is attained. For instance, consider three scores $s_1 = 0$, $s_2 = 1$, and $s_3 = 1$. Then

$$\max_{k \in \{1,2,3\}} s_k = 1, \quad \operatorname{argmax}_{k \in \{1,2,3\}} s_k = \{2,3\}. \quad (32)$$

In (32), the largest score is the number 1, whereas the maximising indices are 2 and 3. If we need to select just one index, we must specify how to break ties. For example, choosing the smallest maximising index gives

$$\min \left(\operatorname{argmax}_{k \in \{1,2,3\}} s_k \right) = 2. \quad (33)$$

The outer $\min$ in (33) acts on the set of indices, not on the scores.

6.2 Several parameters

The same idea extends to objectives with several parameters. Let $\mathcal{L} : \mathbb{R}^d \rightarrow \mathbb{R}$ be an objective whose argument is a column $\boldsymbol{\theta} = (\theta_1, \dots, \theta_d)^\top$. A *local minimiser* $\boldsymbol{\theta}^*$ satisfies

$$\mathcal{L}(\boldsymbol{\theta}^*) \leq \mathcal{L}(\boldsymbol{\theta}^* + \boldsymbol{\Delta}) \quad \text{for every sufficiently small perturbation } \boldsymbol{\Delta}.$$

Fermat's theorem in several variables. If $\mathcal{L}$ is differentiable at a local minimiser $\boldsymbol{\theta}^*$, then

$$\boxed{\nabla \mathcal{L}(\boldsymbol{\theta}^*) = \mathbf{0}}. \quad (34)$$

To obtain (34), fix all coordinates except coordinate j at their values in $\boldsymbol{\theta}^*$. The resulting function of one variable has a local minimum at θ_j^* . Applying the one-variable condition (31) to this function shows that its derivative is zero. That derivative is the j -th component of $\nabla \mathcal{L}(\boldsymbol{\theta}^*)$. Since the argument applies to every coordinate, the entire gradient is zero.

A zero gradient does not certify a (local) minimum. A point at which the gradient is zero is called a *stationary point*. As in one variable, the necessary condition in (34) is not sufficient for a local minimum. For example, consider a column $\mathbf{u} = (u_1, u_2)^\top$ and the function

$$F(\mathbf{u}) = u_1^2 - u_2^2, \quad \nabla F(\mathbf{u}) = \begin{pmatrix} 2u_1 \\ -2u_2 \end{pmatrix}. \quad (35)$$

The function in (35) satisfies $\nabla F(\mathbf{0}) = \mathbf{0}$. Nevertheless, for every $t \neq 0$,

$$F\left(\begin{pmatrix} 0 \\ t \end{pmatrix}\right) = -t^2 < 0 = F(\mathbf{0}), \quad F\left(\begin{pmatrix} t \\ 0 \end{pmatrix}\right) = t^2 > 0 = F(\mathbf{0}).$$

There are therefore both smaller and larger values arbitrarily close to the origin. It is neither a local minimum nor a local maximum; it is a *saddle point*.

In the next lecture, we will apply these optimisation ideas to regression. Later, we will also use gradients to construct an iterative procedure for reducing an objective, called *gradient descent*.

7 Exercises

The exercises practise the tools introduced above. Even if the early questions look familiar, please try Exercise 12: it prepares a calculation we will use in the next lectures. Exercises 13 and 14 practise partial derivatives, gradients and the scalar chain rule with other parameters held fixed. Exercise 15 combines exponentials and logarithms using the chain rule. All dimensions below are positive integers.

1. For $x > 0$, calculate

$$\frac{d}{dx} \left(4x^{5/2} + x^{1/2} - 6x^{-1/2} \right).$$

2. Use the chain rule to calculate

$$\frac{d}{dx} (2x + 5)^3.$$

Name the inner and outer functions and indicate where their derivatives are evaluated.

3. Calculate the same derivative by expanding $(2x + 5)^3$ first. Check that the answers agree.
4. Find the global minimiser and the minimum value of

$$F(a) := 4a^2 + 2a + 1, \quad a \in \mathbb{R}.$$

Justify that the stationary point you find is indeed a global minimiser.

5. Calculate $\sum_{i=0}^5 i^2$.

6. Calculate $\prod_{i=1}^5 i$.

7. Simplify $\sum_{i=1}^N b$, where b does not depend on i .

8. Simplify $\prod_{i=1}^N b$, where b does not depend on i .

9. Compute

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}.$$

Compare the result with the product in Section 3.3, where the order was reversed.

10. Check invertibility and compute the inverse of

$$\mathbf{A} := \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

11. Use the inverse from Exercise 10 to solve

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

Verify your solution by substitution.

12. Let $\mathbf{X} \in \mathbb{R}^{N \times d}$ be fixed and define the vector-valued function

$$\mathbf{g} : \mathbb{R}^d \longrightarrow \mathbb{R}^N, \quad \mathbf{g}(\mathbf{w}) := \mathbf{X}\mathbf{w}.$$

Write $\mathbf{w} = (w_1, \dots, w_d)^\top$. Its i th output component is the scalar function g_i . For $1 \leq i \leq N$ and $1 \leq k \leq d$, find

$$\partial_k g_i(\mathbf{w}) = \left(\frac{\partial g_i}{\partial w_k} \right) (\mathbf{w})$$

in terms of the entries of $\mathbf{X}$. Here $\mathbf{w}$ and $\mathbf{g}(\mathbf{w})$ are columns. Begin by writing $g_i(\mathbf{w})$ as a sum, and explain which quantities remain fixed during differentiation. State whether the derivative you obtain is a scalar, a vector or a matrix.

13. Let

$$F(a, b) := a^2 + ab + 2b^2.$$

Find its two partial derivatives and evaluate them at $(1, 2)$. With a column $\mathbf{w} = (w_1, w_2)^\top$, define $Q(\mathbf{w}) := F(w_1, w_2)$ and give its gradient at $\mathbf{w} = (1, 2)^\top$. State what is held fixed in each partial derivative.

14. Define

$$H(a, b) := (2a - b + 1)^2.$$

Use the scalar chain rule to calculate both partial derivatives. Identify the inner and outer functions, and evaluate the derivatives at $(1, 2)$. Now define

$$K(a, b) := \log(2a - b + 1)$$

on the domain $2a - b + 1 > 0$. Calculate its two partial derivatives using the same inner function.

15. Consider the function

$$F(s) := \log(1 + e^s), \quad s \in \mathbb{R}. \tag{36}$$

Explain why the function in (36) is defined for every real s . Use the chain rule to calculate $F'(s)$, identifying the inner and outer functions and where their derivatives are evaluated. Find $F'(0)$.